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Shao-Peng Yang

Google Scholar
spypaul

Profile

- A Ph.D. student in Electrical and Computer Engineering exploring Memory Systems, Operating Systems, and Storage Systems
- Gaining experience designing and optimizing computer architectures and systems, with a particular interest in CXL technology

Education

Ph.D. in Electrical and Computer Engineering — **Fall 2021 Year - Spring 2027 (expected)**
Syracuse University

Bachelor of Science in Electrical and Computer Engineering — **Fall 2017 - Spring 2021**
Western Washington University — *GPA: 3.96/4.0 | IEEE-Eta Kappa Nu*

Skills

Programming	C++, Parallel and Multithreaded Programming, C, Assembly, Python, Matlab
Hardware Design	Verilog, Altium, Digital Logic Design, skills in utilizing lab tools such as oscilloscopes

Experience

Research Assistant — **Fall 2021 - Present**
Syracuse University
Advisor: Dr. Bryan S. Kim

- **Energy efficient CXL Memory Systems** (Spring 2026 – present)
  - Investigating potential energy inefficiency in CXL memory systems
  - ...
- **Optimizing performance for CXL-based NDP systems** (Spring 2025 – Spring 2026)
  - Studied new execution models for CXL-based NDP systems with peer-to-peer (P2P) communication
  - Proposed near-data processing with collaborative peers (NDP-CP) to break the assumption that NDP devices operate independently
  - Implemented and evaluated the system on a NUMA-based emulation platform
- **CXL integration to LLM training system** (Spring 2024 - Spring 2025)
  - Studied the state-of-the-art LLM training systems and techniques such as 3D parallelism and ZeRO
  - Designed a CXL-based system utilizing memory pooling and sharing to improve the performance and the cost efficiency for LLM training
  - Built a simulator in Python to estimate training performance for a large GPU system with integration of CXL devices
- **Memory management in CXL fabric architecture** (Fall 2023)
  - Explored the design challenges of the CXL fabric manager for achieving high system performance and balancing device loads
  - Researched CXL memory management in terms of allocation, translation, and protection
- **Design of CXL-flash** (Fall 2022 - Spring 2023)
  - Investigated the design challenges for creating a novel memory expansion device using CXL and flash memory
  - Applied techniques such as caching and prefetching to hide the high latency of flash memory in the CXL-flash
  - Built a trace-driven simulator in C++ for testing the CXL-flash design

Intern, Memory Solutions Research Engineer **May 2025 - August 2025**
Samsung Semiconductor, Inc.
Supervisor: Caroline Kahn

- **CMM-H for LLM Inference**
  - Explored the benefits of CMM-H for CPU-based multi-turn LLM inference
  - Studied how CMM-H can store KV cache for long sequences and model weights
- **CMM-H and Data Trimming**
  - Identified a limitation of CMM-H as memory, where data is not invalidated after use
  - Showcased performance overheads due to high write amplification when CMM-H flash memory becomes "full", as stale data is not invalidated

Publications

- **Shao-Peng Yang**, Sandy Lin, Gongjin Sun, Dongyang Li, Shuyi Pei, Bryan Kim. Rethinking Near-Data Processing Execution: Enabling Device Collaboration with CXL Peer-to-Peer Communication. To appear in *IEEE Cluster*, 2026
- Hanqiu Chen, **Shao-Peng Yang (co-first author)**, Mohammadreza Soltaniyeh, Shuyi Pei, Andrew Chang, Bryan S. Kim, Cong Hao. COMETS: Cost-effective Multi-node Efficient Training System with Memory Pooling and Sharing. *ACM International Conference on Supercomputing (ICS)*, 2026
- **Shao-Peng Yang**, Minjae Kim, Sanghyun Nam, Juhyung Park, Jin-yong Choi, Eyee Hyun Nam, Eunji Lee, Sungjin Lee, Bryan S. Kim. Overcoming the Memory Wall with CXL-enabled SSDs. *USENIX Annual Technical Conference (ATC)*, 2023

Course Projects

C++ Programming
- Practiced **multi-threading** techniques for utilizing mutexes, locks, and conditional variables
- Applied **parallel programming** techniques on algorithms such as Fast Fourier Transform and Bitonic Sort
- Implemented various data structures and algorithms such as a red-black tree and searching algorithms
- Practiced advanced concepts such as operator overloading and copy constructors

FPGA Projects with Verilog
- Utilized a **Xilinx Zynq UltraScale+ MPSoC** board for hardware/software co-design
- Implemented a simple **RISC-V** processor in Verilog

Internet Security Labs from SEED LABS
- Worked on labs in a simulated environment to understand network and security details at network layers
- Mainly utilized **Python with Scapy** and tools such as **Wireshark/tcpdump**

Relevant Courses

Object Oriented Programming in C++, Parallel Programming and Multi-Threading, Advanced Computer Architecture, Introduction to System-on-Chip Design, Internet Security